

SAIRAM CHAKRALA

Software Engineer • Backend Systems & Distributed Architecture

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EXPERIENCE

Software Engineer

Enphase Energy

07/2024 - Present | India

A leading provider of energy management technology delivering solar energy solutions and EV cloud infrastructure.

- Microservice Ownership:** Independently planned, implemented, and maintained 3 critical microservices within a 10+ service cloud platform serving 15,000+ EV chargers globally — covering charging session management, telemetry ingestion, and fleet analytics, collectively handling 20M+ events daily. Followed agile development practices with full unit test coverage and peer code reviews for every feature.
- AI-Assisted Development Workflows:** Pioneered AI adoption within the engineering team, standardising 5 LLD templates and documentation processes across 3 service domains — boosting team productivity by ~30% and reducing code review turnaround from 2 days to under 1 day by enabling AI-assisted review pipelines.
- Bidirectional Charger Support (V2G):** Led core backend implementation for newly launched bidirectional (V2G) charger features, developing session state machines and OCPP 1.6-compliant telemetry across 3+ partner charger types — reducing partner integration effort by ~40% by standardising the session lifecycle API.
- Green Energy Scheduling & Cost Optimisation:** Created an intelligent energy management system aligning EV charging cycles with peak solar production — achieving 50-60% energy cost reduction for fleet operators while maintaining 99.9% data reliability.
- Fleet Analytics Dashboard (Push-Based Architecture):** Built an Analytics Service (Java/Spring Boot + Go/Lambda) aggregating data from 15K+ chargers into MongoDB, tracking 5M+ sessions. Adopted a push-based architecture eliminating polling — enabling sub-second latency updates for operators monitoring the full fleet.
- High-Throughput Telemetry Pipeline (Kafka · MongoDB):** Owned end-to-end design of a telemetry and network-stats pipeline processing 20M+ daily OCPP 1.6 Meter Value reports across 15K+ chargers. Persisted time-series data enabling EV installers to detect charger anomalies in real time — cutting mean diagnostic time by ~40%.
- Backward-Compatible Schema Evolution:** Developed dual-resolution telemetry (1-min & 5-min) with versioned Kafka message envelopes — achieving zero downtime migration across 5+ dependent services and 15K+ chargers in production.
- Real-Time IoT Livestreaming (AWS IoT Core):** Built a real-time telemetry livestream via AWS Lambda and IoT Core, providing operators a live view of 10,000+ concurrent sessions with sub-second (<1s) latency and zero polling overhead.
- Performance Engineering & Caching (Redis):** Reduced inter-service API calls from 1M/day to under 5K/day (99.5%) and cut average latency from ~300ms to ~120ms during peak load — improving scalability, resilience, and maintainability.
- Cross-Functional Collaboration & On-Call:** Led a cross-team initiative across Backend, Firmware, and DevOps to redesign the CI/CD pipeline — reducing deployment time by ~30%. Actively handled on-call responsibilities and led RCA documentation to drive long-term reliability.

SUMMARY

Backend engineer with 2 years of production experience building distributed, high-throughput systems at Enphase Energy. Planned and owned end-to-end microservices — from Kafka-based telemetry pipelines ingesting 20M+ daily events to bidirectional (V2G) charger backends and charge session management systems serving a fleet of 15,000+ EV chargers globally. Proficient in Java and Go; experienced with object-oriented design, design patterns, low-level design (LLD), and agile development. IIT Roorkee CS graduate (AIR 1249 JEE Advanced 2020) with a strong problem-solving foundation (Codeforces Expert, rating 1645).

EDUCATION

B.Tech in Computer Science

IIT Roorkee

08/2020 - 05/2024 | GPA: 8.27 / 10

Intermediate (MPC)

Narayana Junior College

2018 - 2020 | Percentage: 92.6%

KEY ACHIEVEMENTS

Competitive Programming

Codeforces Expert — Rating 1645. Ranked in the top ~11% globally among 300,000+ active competitors. Consistent high performer in rated algorithmic contests.

JEE Advanced AIR 1249 / JEE Mains AIR 874

Top-0.1% percentile among 1.5M+ candidates in India's most competitive engineering entrance exam.

SKILLS

Java	Spring Boot	Go
Kafka	AWS Kinesis	MongoDB
DocumentDB	Redis	AWS Lambda
AWS IoT Core	Microservices	Multi-threading
System Design	LLD	Datadog

Git